Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 47: Guess an Exponential

Solving Linear Recurrences

MCS Section 21.3, pp. 874–883

Name: _____ Date: _____

Where we are

Monday produced equations and conjectures. Today produces *answers* — a procedure that derives the closed form instead of guessing it.
The Fibonacci recurrence went unsolved for almost six centuries. The method below takes about an hour to learn.

By the end of today you should be able to

1. Write and solve the characteristic equation of a homogeneous linear recurrence.
2. Handle a repeated root.
3. Solve an inhomogeneous recurrence by the five-step method.

1. Homogeneous linear recurrences

Definition

A **homogeneous linear recurrence** has the form

$$f(n) = a_1 f(n-1) + a_2 f(n-2) + \cdots + a_d f(n-d)$$

with constants a_i. The number d is its **order**.

The idea. Linear recurrences tend to have exponential solutions, so guess $f(n) = x^n$ and find which x work. For Fibonacci, $f(n) = f(n-1) + f(n-2)$ gives

$$x^n = x^{n-1} + x^{n-2}.$$

Dividing by x^{n-2} leaves _____, a quadratic.

The characteristic equation

For $f(n) = a_1 f(n-1) + \cdots + a_d f(n-d)$, the characteristic equation is

$$x^d = a_1 x^{d-1} + a_2 x^{d-2} + \cdots + a_d.$$

Its coefficients are read straight off the recurrence.

Theorem 21.3.1

If f and g both solve a homogeneous linear recurrence, so does $sf + tg$ for any reals s, t.

So each root r contributes a solution r^n, and the general solution is a _____ of them. The boundary conditions then pin down the constants.

2. Worked: distinct roots

Solve $f(n) = 5f(n-1) - 6f(n-2)$ with $f(0) = 1$, $f(1) = 0$.

Step 1. Characteristic equation: _____, that is $x^2 - 5x + 6 = 0$, with roots $x =$ _____ and $x =$ _____.

Step 2. General solution: $f(n) = c \cdot 2^n + d \cdot 3^n$.

Step 3. Apply the boundary conditions:

$$f(0) = c + d = 1, \qquad f(1) = 2c + 3d = 0.$$

Solving: $c =$ _____, $d =$ _____.

$$f(n) = 3 \cdot 2^n - 2 \cdot 3^n$$

Check $n = 2$: formula gives $12 - 18 = -6$; recurrence gives $5(0) - 6(1) =$ _____. ✓

Fibonacci, for the record

The same method on $x^2 = x + 1$ gives roots $\frac{1 \pm \sqrt{5}}{2}$ and, after the boundary conditions, **Binet's formula**:

$$f(n) = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

Every Fibonacci number is an integer, yet the formula is full of $\sqrt{5}$ — the irrational parts cancel exactly. The larger root $\frac{1+\sqrt{5}}{2} \approx 1.618$ is the golden ratio, and the second term shrinks to nothing, which is why consecutive Fibonacci numbers approach that ratio.

3. Repeated roots

The rule

A root r of multiplicity k contributes k solutions:

$$r^n, \quad nr^n, \quad n^2 r^n, \ \ldots, \ n^{k-1} r^n.$$

Solve $f(n) = 4f(n-1) - 4f(n-2)$ with $f(0) = 1$, $f(1) = 4$.

Characteristic: $x^2 - 4x + 4 = (x-2)^2$, so $r =$ _____, twice.

General solution: $f(n) = c \cdot 2^n + d \cdot n2^n$.

$f(0) = c = 1$; $\quad f(1) = 2c + 2d = 4$ gives $d =$ _____.

$$f(n) = 2^n + n2^n$$

Check $n = 3$: $8 + 24 = 32$, and the recurrence gives $4(12) - 4(4) =$ _____. ✓

Why nr^n and not a second r^n

With a repeated root there is only *one* exponential, so $cr^n + dr^n$ collapses to a single constant times r^n — not enough freedom to meet two boundary conditions. The factor of n supplies the missing degree of freedom.

4. Inhomogeneous: the five-step method

An extra term $g(n)$ on the right makes the recurrence **inhomogeneous**: $f(n) = a_1 f(n-1) + \cdots + g(n)$.

The procedure

1. Drop $g(n)$ and find the roots of the characteristic equation.
2. Write the **homogeneous solution**, constants undetermined.
3. Find *any* **particular solution** to the full recurrence, ignoring the boundary conditions. Guess a form matching $g(n)$.
4. Add the two to get the **general solution**.
5. Now apply the boundary conditions to fix the constants.

Guessing a particular solution. If $g(n)$ is constant, try $f(n) = c$. If $g(n)$ is a polynomial, try one of the same degree. If $g(n) = 3^n$, try $c \cdot 3^n$.

Worked (MCS). A Hanoi variant where moving disk k takes k seconds: $f(1) = 1$, $f(n) = 2f(n-1) + n$.

Steps 1–2. Homogeneous part $f(n) = 2f(n-1)$ has characteristic $x = 2$, so the homogeneous solution is _____.

Step 3. Since $g(n) = n$ is degree 1, guess $f(n) = an + b$. Substituting:

$$an + b = 2\big(a(n-1) + b\big) + n \implies 0 = (a+1)n + (b - 2a).$$

This holds for all n when $a =$ _____ and $b =$ _____. So $-n - 2$ is a particular solution.

Step 4. General: $f(n) = c \cdot 2^n - n - 2$.

Step 5. $f(1) = 2c - 3 = 1$ gives $c =$ _____.

$$f(n) = 2 \cdot 2^n - n - 2$$

What the answer says

The original puzzle cost $2^n - 1$. Charging by disk size costs about _____ as much — not exponentially more. Making each step more expensive barely matters; it is the *number* of subproblems that drives the cost.
That observation is Friday's whole subject.

Exit ticket

1. Write the characteristic equation of $f(n) = f(n-1) + 6f(n-2)$ and find its roots.

2. Why does a repeated root need the extra solution nr^n?

Before next class

Read *MCS* Section 21.4, "Divide-and-Conquer Recurrences," and Section 21.5.

Questions I still have